Bài 1:

* Bật cJSON trong buildroot và build lại OS tích hợp thư viện JSON:

- Trong thư mục buildroot, chạy lệnh "make menuconfig".
- Bật cJSON:

```
/home/son/buildroot/.config - Buildroot 2023.02-dirty Configuration
→ Target packages → Libraries → JSON/XML
                                   JSON/XML
  Arrow keys navigate the menu.  <Enter> selects submenus ---> (or empty submenus ----).  Highlighted letters are hotkeys.  Pressing <Y> selects a feature, while <N> excludes a
  feature.  Press <Esc><Esc> to exit, <?> for Help, </> for Search.  Legend: [*] feature is selected  [ ] feature is excluded

                                       *** benejson needs a toolchain w/ C++ ***
                                   [*] cJSON
                                   [ ]   cJSON utils (NEW)
                                   [ ] expat
                                   [ ] jansson
                                   [ ] jose
                                   [ ] jsmn
                                   [ ] json-c
                                       *** json-for-modern-cpp needs a toolchain w/ C++, gcc >= 4.9 ***
                                   [ ] json-glib
                                       *** jsoncpp needs a toolchain w/ C++, gcc >= 4.7 ***
                                   [ ] libbson
                                   [ ] libfastjson
                                       *** libjson needs a toolchain w/ C++ ***
                                   [ ] roxml
                                   [ ] libucl
                                   [ ] libxml2
                                       *** libxml++ needs a toolchain w/ C++, wchar, threads, gcc >= 7 ***
                                   [ ] libxmlrpc
                                   [ ] libxslt
                                   [ ] libyaml
                                   [ ] Mini-XML
                                       *** pugixml needs a toolchain w/ C++ ***
                                       *** rapidjson needs a toolchain w/ C++ ***
                                   [ ] rapidxml
                                   [ ] raptor
                                   [ ] serd
                                   [ ] sord
                                       *** tinyxml needs a toolchain w/ C++ ***
                                       *** tinyxml2 needs a toolchain w/ C++ ***
                                       *** valijson needs a toolchain w/ C++ ***
                                   ↓(+)

                                        <Select>    < Exit >    < Help >    < Save >    < Load >
```

- Dùng lệnh "make” để build lại.
- Build lại OS:

```
son@son-Precision-5540:~$ cd ~/buildroot
son@son-Precision-5540:~/buildroot$ make menuconfig
configuration written to /home/son/buildroot/.config

*** End of the configuration.
*** Execute 'make' to start the build or try 'make help'.

son@son-Precision-5540:~/buildroot$ make
/usr/bin/make -j1 O=/home/son/buildroot/output HOSTCC="/usr/bin/gcc" HOSTCXX="/usr/bin/g++" syncconfig
>>> host-cmake 3.22.3 Downloading
wget --passive-ftp -nd -t 3 -O '/home/son/buildroot/output/build/.cmake-3.22.3.tar.gz.zHbf5l/output' 'https://cmake.org/files/v3.22/cmake-3.22.3.tar.gz'
--2026-03-14 20:10:38--  https://cmake.org/files/v3.22/cmake-3.22.3.tar.gz
Resolving cmake.org (cmake.org)... 66.194.253.25
Connecting to cmake.org (cmake.org)|66.194.253.25|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9779118 (9.3M) [application/x-gzip]
Saving to: ‘/home/son/buildroot/output/build/.cmake-3.22.3.tar.gz.zHbf5l/output’

/home/son/buildroot/output/build/.cmake-3.22. 100%[===================================================================================================>]   9.33M  2.31MB/s    in 6.0s

2026-03-14 20:10:45 (1.55 MB/s) - ‘/home/son/buildroot/output/build/.cmake-3.22.3.tar.gz.zHbf5l/output’ saved [9779118/9779118]

cmake-3.22.3.tar.gz: OK (sha256: 9f8469166f94553b6978a16ee29227ec49a2eb5ceb608275dec40d8ae0d1b5a0)
>>> host-cmake 3.22.3 Extracting
gzip -d -c /home/son/buildroot/dl/cmake/cmake-3.22.3.tar.gz | tar --strip-components=1 -C /home/son/buildroot/output/build/host-cmake-3.22.3   -xf -
>>> host-cmake 3.22.3 Patching

Applying 0001-rename-cmake-rootfile.patch using patch:
patching file Source/cmake.cxx
Hunk #1 succeeded at 1527 with fuzz 1 (offset 518 lines).
>>> host-cmake 3.22.3 Configuring
(cd /home/son/buildroot/output/build/host-cmake-3.22.3; PATH="/home/son/buildroot/output/host/bin:/home/son/buildroot/output/host/sbin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin" PKG_CONFIG="/home/son/buildroot/output/host/bin/pkg-config" PKG_CONFIG_SYSROOT_DIR="/" PKG_CONFIG_ALLOW_SYSTEM_CFLAGS=1 PKG
_CONFIG_ALLOW_SYSTEM_LIBS=1 PKG_CONFIG_LIBDIR="/home/son/buildroot/output/host/lib/pkgconfig:/home/son/buildroot/output/host/share/pkgconfig" AR="/usr/bin/ar" AS="/usr/bin/as" LD="/usr
/bin/ld" NM="/usr/bin/nm" CC="/usr/bin/gcc" GCC="/usr/bin/gcc" CXX="/usr/bin/g++" CPP="/usr/bin/cpp" OBJCOPY="/usr/bin/objcopy" RANLIB="/usr/bin/ranlib" CPPFLAGS="-I/home/son/buildroot
/output/host/include" CFLAGS="-O2 -I/home/son/buildroot/output/host/include" CXXFLAGS="-O2 -I/home/son/buildroot/output/host/include" LDFLAGS="-L/home/son/buildroot/output/host/lib -Wl
,-rpath,/home/son/buildroot/output/host/lib" INTLTOOL_PERL=/usr/bin/perl CFLAGS="-O2 " ./bootstrap --prefix=/home/son/buildroot/output/host --parallel=13 -- -DCMAKE_C_FLAGS="-O2 " -DCM
AKE_CXX_FLAGS="-O2 " -DCMAKE_EXE_LINKER_FLAGS="-L/home/son/buildroot/output/host/lib -Wl,-rpath,/home/son/buildroot/output/host/lib" -DCMAKE_USE_OPENSSL:BOOL=OFF -DBUILD_CursesDialog=O
FF )
------------------------------------------
CMake 3.22.3, Copyright 2000-2021 Kitware, Inc. and Contributors
C compiler on this system is: /usr/bin/gcc -O2
C++ compiler on this system is: /usr/bin/g++ -O2 -I/home/son/buildroot/output/host/include
Makefile processor on this system is: gmake
/usr/bin/g++ has setenv
```

* Viết ct C sử dụng thư viện CJSON:

```
  GNU nano 7.2                                                    HelloJSON.c
#include <stdio.h>
#include <stdlib.h>
#include "cJSON.h"

int main()
{
    char *json_string = "{\"name\":\"BBB\",\"year\":2026,\"status\":\"OK\"}";

    cJSON *json = cJSON_Parse(json_string);

    if (json == NULL)
    {
        printf("Parse error\n");
        return -1;
    }

    cJSON *name = cJSON_GetObjectItem(json, "name");
    cJSON *year = cJSON_GetObjectItem(json, "year");
    cJSON *status = cJSON_GetObjectItem(json, "status");

    printf("Name: %s\n", name->valuestring);
    printf("Year: %d\n", year->valueint);
    printf("Status: %s\n", status->valuestring);

    cJSON_Delete(json);

    return 0;
}

                                                  [ Read 28 lines ]
^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket  M-Q Previous   ^B Back
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^Q Where Was    M-W Next       ^F Forward
```

* Thực hiện biên dịch chéo:

```
son@son-Precision-5540:~/buildroot$ nano HelloJSON.c
son@son-Precision-5540:~/buildroot$ nano HelloJSON.c
son@son-Precision-5540:~/buildroot$ export PATH=$PATH:~/buildroot/output/host/bin
son@son-Precision-5540:~/buildroot$ arm-linux-gcc HelloJSON.c -o HelloJSON -lcjson
HelloJSON.c:3:10: fatal error: cJSON.h: No such file or directory
    3 | #include "cJSON.h"
      |          ^~~~~~~~
compilation terminated.
son@son-Precision-5540:~/buildroot$ arm-linux-gcc HelloJSON.c \
-I output/staging/usr/include \
-L output/staging/usr/lib \
-lcjson \
-o HelloJSON
HelloJSON.c:3:10: fatal error: cJSON.h: No such file or directory
    3 | #include "cJSON.h"
      |          ^~~~~~~~
compilation terminated.
son@son-Precision-5540:~/buildroot$ ls output/staging/usr/include | grep -i json
cjson
son@son-Precision-5540:~/buildroot$ ls output/staging/usr/include/cjson
cJSON.h
son@son-Precision-5540:~/buildroot$ nano HelloJSON.c
son@son-Precision-5540:~/buildroot$ arm-linux-gcc HelloJSON.c \
-I output/staging/usr/include/cjson \
-L output/staging/usr/lib \
-lcjson \
-o HelloJSON
son@son-Precision-5540:~/buildroot$ file HelloJSON
HelloJSON: ELF 32-bit LSB pie executable, ARM, EABI5 version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-armhf.so.3, for GNU/Linux 5.10.0, not stripped
son@son-Precision-5540:~/buildroot$
```

* Copy ct sang bbb:

- Copy vào thẻ sd: sudo cp HelloJSON /media/son/rootfs1/
- Flash hệ điều hành xuống thẻ sd: sudo cp output/target/usr/lib/libcjson.so* /media/son/rootfs1/lib/
- Chạy thành công chương trình trên BBB:

```
Welcome to Buildroot
buildroot login: root
# /HelloJSON
Name: BBB
Year: 2026
Status: OK
#
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.9 | VT102 | Offline | ttyUSB0
```

Bài 2:

- Tạo thư mục:

```
son@son-Precision-5540:~/buildroot$ mkdir mylib
son@son-Precision-5540:~/buildroot$ cd mylib
son@son-Precision-5540:~/buildroot/mylib$
```

- Tạo file header: vi mylib.h
- Tạo file source: vi mylib.c
- Dùng lệnh "arm-linux-gcc -c mylib.c" để biên dịch thư viện tạo ra file mylib.o
- Dùng lệnh "arm-linux-ar rcs libmylib.a mylib.o" để tạo ra libmylib.a (Static Library).
- Dùng lệnh "arm-linux-gcc -shared -fPIC mylib.c -o libmylib.so" để tạo ra libmylib.so (Dynamic Library).
- Ném thư viện vào sysroot:

  cp mylib.h ~/buildroot/output/host/arm-buildroot-linux-gnueabihf/sysroot/usr/include/
  cp libmylib.a ~/buildroot/output/host/arm-buildroot-linux-gnueabihf/sysroot/usr/lib/
  cp libmylib.so ~/buildroot/output/host/arm-buildroot-linux-gnueabihf/sysroot/usr/lib/
- Viết ct sử dụng thư viện : vi main.c
- Biên dịch ct:

Static:

```bash
arm-linux-gcc main.c -static -lmylib -o app_static
```

Dynamic:

```bash
arm-linux-gcc main.c -lmylib -o app_dynamic
```

- Cắm thẻ SD vào và copy ct vào thẻ sd:

```
son@son-Precision-5540:~/buildroot/mylib$ sudo cp app_static /media/son/rootfs1/
[sudo] password for son:
son@son-Precision-5540:~/buildroot/mylib$ sudo cp app_dynamic /media/son/rootfs1/
son@son-Precision-5540:~/buildroot/mylib$ sudo cp libmylib.so /media/son/rootfs1/lib/
son@son-Precision-5540:~/buildroot/mylib$ sudo umount /media/son/rootfs1
son@son-Precision-5540:~/buildroot/mylib$
```

- Rút thẻ SD cắm vào BBB, mở nguồn, mở uart rồi chạy ct:

```
Welcome to Buildroot
buildroot login: root
# /app_static
Add: 13
Max: 8
# /app_dynamic
Add: 13
Max: 8
#
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.9 | VT102 | Offline | ttyUSB0
```

- So sánh kích thước của 2 ct: app static lớn hơn vì thư viện đã được nhúng vào ct.

```
son@son-Precision-5540:~/buildroot/mylib$ ls -lh app_static app_dynamic
-rwxrwxr-x 1 son son 7.7K Mar 14 22:29 app_dynamic
-rwxrwxr-x 1 son son 534K Mar 14 22:29 app_static
son@son-Precision-5540:~/buildroot/mylib$
```

- Kiểm tra dependency bằng `readelf:`

`app_dynamic: CT phụ thuộc thư viện động.`

```
son@son-Precision-5540:~/buildroot/mylib$ readelf -d app_dynamic

Dynamic section at offset 0xebc contains 29 entries:
  Tag        Type                         Name/Value
 0x00000001 (NEEDED)                     Shared library: [libmylib.so]
 0x00000001 (NEEDED)                     Shared library: [libc.so.6]
 0x0000000c (INIT)                       0x414
 0x0000000d (FINI)                       0x66c
 0x00000019 (INIT_ARRAY)                 0x10eb4
 0x0000001b (INIT_ARRAYSZ)               4 (bytes)
 0x0000001a (FINI_ARRAY)                 0x10eb8
 0x0000001c (FINI_ARRAYSZ)               4 (bytes)
 0x00000004 (HASH)                       0x190
 0x6ffffef5 (GNU_HASH)                   0x1d4
 0x00000005 (STRTAB)                     0x2ac
 0x00000006 (SYMTAB)                     0x1ec
 0x0000000a (STRSZ)                      167 (bytes)
 0x0000000b (SYMENT)                     16 (bytes)
 0x00000015 (DEBUG)                      0x0
 0x00000003 (PLTGOT)                     0x10fc4
 0x00000002 (PLTRELSZ)                   56 (bytes)
 0x00000014 (PLTREL)                     REL
 0x00000017 (JMPREL)                     0x3dc
 0x00000011 (REL)                        0x39c
 0x00000012 (RELSZ)                      64 (bytes)
 0x00000013 (RELENT)                     8 (bytes)
 0x00000018 (BIND_NOW)
 0x6ffffffb (FLAGS_1)                    Flags: NOW PIE
 0x6ffffffe (VERNEED)                    0x36c
 0x6fffffff (VERNEEDNUM)                 1
 0x6ffffff0 (VERSYM)                     0x354
 0x6ffffffa (RELCOUNT)                   4
 0x00000000 (NULL)                       0x0
```

`app_static: không phụ thuộc thư viện ngoài.`

```
son@son-Precision-5540:~/buildroot/mylib$ readelf -d app_static

There is no dynamic section in this file.
son@son-Precision-5540:~/buildroot/mylib$
```

Bài 3:

3.1.Đưa thư viện bài 2 vào buildroot.

- Vào thư mục package của Buildroot: cd ~/buildroot/package
- Tạo package: mkdir mylib
  cd mylib
- Tạo file Config_in bằng lệnh vi.
- Tạo file mylib.mk bằng lệnh vi.

3.2. Viết CT dùng 2 thư viện.

- Tạo CT myapp.c bằng nano myapp.c

3.3. Biện dịch và chạy trên bbb

- Biên dịch: ~/buildroot/output/host/bin/arm-buildroot-linux-gnueabihf-gcc myapp.c -o myapp -lcjson -lmylib
- Copy vào thẻ sd và đưa xuống bbb.
- Chạy myapp trên BBB:

```
Welcome to Buildroot
buildroot login: root
# ./myapp
-sh: ./myapp: not found
# myapp
Add: 12
Max: 7
{
        "board":        "BBB"
}
#
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.9 | VT102 | Offline | ttyUSB0
```

3.4. Tạo package cho ứng dụng.

- Đi vào thư mục package: cd ~/buildroot/package
- Tạo thư mục myapp:
  mkdir myapp
  cd myapp
- Tạo file Config_in : vi Config_in.
- Tạo file myapp.mk.
- Thêm package vào buildroot:

//mở vi ~/buildroot/package/Config.in

//thêm dòng source "package/myapp/Config.in"

- Flash xuông bbb và chạy ct.

```
Welcome to Buildroot
buildroot login: root
# ./myapp
-sh: ./myapp: not found
# myapp
Add: 12
Max: 7
{
        "board":        "BBB"
}
#
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.9 | VT102 | Offline | ttyUSB0
```